

Quantum Information Theory

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Abstract

These are lecture notes for the course NMMB537 Quantum Information Theory, taught at Faculty of Mathematics and Physics, Charles University.

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Chapter 1

Quantum States and Measurements

We start by describing a precise mathematical framework for quantum states.

1.1 Finite-dimensional Hilbert Spaces

Axiom 1 (Hilbert space). To every quantum system we associate a Hilbert space $\mathcal{H}$.

We only consider finite-dimensional Hilbert spaces in this course. If $\dim(\mathcal{H}) = d$, then $\mathcal{H}$ is isomorphic to

$$\mathbb{C}^d = \left\{ \begin{pmatrix} \psi_1 \\ \vdots \\ \psi_d \end{pmatrix} : \psi_i \in \mathbb{C} \right\}$$

with the standard inner product

$$\langle \psi | \phi \rangle = \sum_{i=1}^d \overline{\psi_i} \phi_i.$$

Bra-ket notation. We write $|\psi\rangle \in \mathcal{H}$ for any vector. By $\langle \psi| \in \mathcal{H}^\dagger$, we mean the linear map $\mathcal{H} \rightarrow \mathbb{C}$ defined by $\langle \psi|(|\phi\rangle) = \langle \psi|\phi\rangle$. In other words, $\langle \psi|$ is the row vector corresponding to the conjugate transpose of $|\psi\rangle$. We denote the standard basis of $\mathbb{C}^d$ by $|0\rangle, |1\rangle, \dots, |d\rangle$. That is, $|i\rangle$ has 1 on the i^{th} coordinate and 0 elsewhere.

Linear algebra notation. Every basis that we will consider will be orthogonal. If there is no confusion, we will use the terms linear map and matrix interchangeably.

By $\text{Lin}(\mathcal{H}, \mathcal{K})$, we denote the set of linear maps $\mathcal{H} \rightarrow \mathcal{K}$. When we fix the standard bases, we can express $M \in \text{Lin}(\mathcal{H}, \mathcal{K})$ as

$$M = \sum_{i,j} M_{ij} |i\rangle\langle j|.$$

Note that $\langle i|M|j\rangle = M_{ij} \in \mathbb{C}$.

Exercise 1.1.1. Write $\sum_{i,j \in \{0,1\}} ij|i\rangle\langle j|$ as a matrix.

If $M \in \text{Lin}(\mathcal{H}, \mathcal{H}) = \text{Lin}(\mathcal{H})$, then the *trace* with respect to the standard basis is

$$\text{tr}[M] = \sum_i M_{ii} = \sum_i \langle i|M|i \rangle.$$

An important fact about trace is that it is in fact independent of the basis. Recall also that $\text{tr}[MN] = \text{tr}[NM]$.

Here are some important types of matrices:

- $M \in \text{Lin}(\mathcal{H})$ is *Hermitian* if $M^\dagger = M$.
- $P \in \text{Lin}(\mathcal{H})$ is *positive* if $\langle \psi|P|\psi \rangle \geq 0$, for all $|\psi \rangle \in \mathcal{H}$. $\text{PSD}(\mathcal{H})$ is the class of all positive matrices. We also use the notation $P \geq 0$ to indicate that P is positive. If $M, N \in \text{Lin}(\mathcal{H})$, then we define $M \geq N$ if $M - N \geq 0$, that is, if $M - N$ is positive.
- $U \in \text{Lin}(\mathcal{H})$ is *unitary* if $U^\dagger U = UU^\dagger = I$. $\mathcal{U}(\mathcal{H})$ is the class of all unitary matrices.
- $V \in \text{Lin}(\mathcal{H}, \mathcal{K})$ is an *isometry* if $V^\dagger V = I$.
- $P \in \text{Lin}(\mathcal{H})$ is a *projection* if $P^\dagger = P$ and $P^2 = P$.

Exercise 1.1.2. Are the following matrices {Hermitian, PSD, unitary, projections}?

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \begin{pmatrix} e^{i\frac{\pi}{4}} & 0 \\ 0 & e^{-i\frac{\pi}{4}} \end{pmatrix}.$$

Theorem 1.1.3 (Spectral theorem for Hermitian matrices). *If $M \in \text{Lin}(\mathcal{H})$ is Hermitian $d \times d$ matrix, then M has eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$ and there is a basis of eigenvectors $\{|\psi_i \rangle\}_{i=1}^d$ such that*

$$M = \sum_{i=1}^d \lambda_i |\psi_i \rangle \langle \psi_i|.$$

Recall that the eigenvalues can have multiple occurrences and the basis of eigenvectors may not be unique. Indeed, for instance,

$$I = \sum_{i=1}^d |\psi_i \rangle \langle \psi_i|$$

is true for any basis $\{|\psi_i \rangle\}_{i=1}^d$.

This allows us to define functions of Hermitian matrices. If $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function, then we can define

$$f(M) = \sum_{i=1}^d f(\lambda_i) |\psi_i \rangle \langle \psi_i|.$$

For instance,

$$\sqrt{M} = \sum_{i=1}^d \sqrt{\lambda_i} |\psi_i \rangle \langle \psi_i|.$$

Note that if P is a projection, then the condition $P^2 = P$ implies that the eigenvalues are in $\{0, 1\}$. Thus, we can write $P = \sum_{i=1}^r |\psi_i \rangle \langle \psi_i|$. Here, $r = \text{rank}(P)$ of the projection P .

Theorem 1.1.4 (Characterization of positive matrices). *If $P \in \text{Lin}(\mathcal{H})$, then the following are equivalent:*

- (a) P is positive, i.e., $\langle \psi | P | \psi \rangle \geq 0$, for all $|\psi\rangle \in \mathcal{H}$.
- (b) $P^\dagger = P$ and all eigenvalues are nonnegative.
- (c) There is $M \in \text{Lin}(\mathcal{H}, \mathcal{K})$ such that $P = M^\dagger M$.
- (d) $\text{tr}[PQ] \geq 0$, for every $Q \in \text{PSD}(\mathcal{H})$.

Exercise 1.1.5. Prove that if $P \in \text{PSD}(\mathcal{H})$ and $M \in \text{Lin}(\mathcal{H}, \mathcal{K})$, then $MPM^\dagger$ is positive.

1.2 Quantum States

Definition 1.2.1. A *density matrix* is a positive operator $\rho \in \text{PSD}(\mathcal{H})$ with $\text{tr}[\rho] = 1$. We put $S(\mathcal{H}) = \{\rho \in \text{PSD}(\mathcal{H}) : \text{tr}[\rho] = 1\}$.

Axiom 2. The state of a quantum system with Hilbert space $\mathcal{H}$ is described by a density matrix ρ . We will refer to ρ as a quantum state.

Example 1.2.2 (classical states). In probability theory we have a finite set of outcomes Σ with a probability distribution p such that $p(x) \geq 0$, for $x \in \Sigma$, and $\sum_{x \in \Sigma} p(x) = 1$. If $\mathcal{H}$ has a basis $\{|x\rangle\}_{x \in \Sigma}$, then

$$\rho = \sum_{x \in \Sigma} |x\rangle\langle x|$$

is a density matrix. Conversely, any diagonal density matrix corresponds to a probability distribution. $\square$

Example 1.2.3 (pure states). Let $|\psi\rangle \in \mathcal{H}$ be a unit vector. Then $\rho = |\psi\rangle\langle\psi|$ is a density matrix since $\text{tr}[|\psi\rangle\langle\psi|] = \text{tr}[\langle\psi|\psi\rangle] = 1$. If $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, then

$$|\psi\rangle\langle\psi| = \begin{pmatrix} |\alpha|^2 & \alpha\bar{\beta} \\ \bar{\alpha}\beta & |\beta|^2 \end{pmatrix}.$$

Note that replacing $|\psi\rangle$ by $e^{i\theta}|\psi\rangle$ does not change the density matrix. $\square$

Exercise 1.2.4. Is $\rho = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ a pure state?

Solution. Yes, since $\rho = |+\rangle\langle+|$, where $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. $\square$

Example 1.2.5 (general state). Let ρ be a general quantum state. By Theorem 1.1.4, we can write

$$\rho = \sum_{i=1}^d p_i |\psi_i\rangle\langle\psi_i|,$$

where $p_i \geq 0$. Since ρ is diagonal in the basis $\{|\psi_i\rangle\}_{i=1}^d$ and trace is independent of the chosen basis, $\text{tr}[\rho] = 1$ implies that $\sum_{i=1}^d p_i = 1$. So, the eigenvalues p_i 's define a probability distribution. We may interpret ρ as if we have the pure state $|\psi_i\rangle\langle\psi_i|$ with probability p_i . We call ρ a *mixed state* if it is not pure.

However, note that the decomposition of ρ given above is not unique. It is possible that ρ may be written as a sum over some different probabilities and different states, even different number of states. $\square$

Exercise 1.2.6. What is the state corresponding to 50% chance of $|0\rangle$ and 50% chance of $|+\rangle$?

Solution.

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|+\rangle\langle +| = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}.$$

Note that the eigenvalues of ρ are $1/2 + \sqrt{5}/4$ and $1/2 - \sqrt{5}/4$, so the spectral decomposition of ρ gives a completely different decomposition of ρ compared to the one we started with. $\square$

Definition 1.2.7. The state $\tau = \frac{1}{d}I$ is called *the maximally mixed state*.

1.3 Bloch Sphere

How does the set $S(\mathcal{H})$ look like? For two states $\rho_1, \rho_2 \in S(\mathcal{H})$ and $t \in [0, 1]$, the convex combination $t\rho_1 + (1-t)\rho_2$ is also a state. In fact, we have the following lemma.

Lemma 1.3.1. *The set $S(\mathcal{H}) \subseteq \text{Lin}(\mathcal{H})$ is convex. Moreover, the extreme points of $S(\mathcal{H})$ are exactly the pure states.*

For the particular case of one qubit, there is a visualization known as the *Bloch sphere*. Hermitian 2×2 matrices have the following (real) basis:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The last three matrices are known as *Pauli matrices*. Here are some of their properties:

$$X^2 = Y^2 = Z^2 = I, \quad XY = iZ = -YX, \quad \text{tr}[X] = \text{tr}[Y] = \text{tr}[Z] = 0.$$

In particular, Pauli matrices are unitary and have eigenvalues ± 1 .

Every 2×2 Hermitian matrix can be written in a unique way as a linear combination of the matrices I, X, Y, Z with all coefficients being real numbers. An arbitrary 2×2 Hermitian matrix ρ with $\text{tr}[\rho] = 1$ can be expressed as

$$\rho = \frac{1}{2}(I + xX + yY + zZ) = \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix}. \quad (1.3.1)$$

The coefficient $1/2$ in front of the I is fixed by the condition $\text{tr}[\rho] = 1$. Further, $\text{tr}[\rho] = 1$ implies that the eigenvalues of ρ are λ and $1 - \lambda$, for some $\lambda \in \mathbb{R}$. If we want ρ to be a quantum state, it also has to be positive, which is equivalent to requiring that $\lambda \geq 0$ and $1 - \lambda \geq 0$. Now, λ and $1 - \lambda$ are nonnegative if and only if $\det(\rho) = \lambda(1 - \lambda) \geq 0$. We compute

$$\det(\rho) = \frac{1}{4}((1+z)(1-z) - (x+iy)(x-iy)) = \frac{1}{4}(1 - x^2 - y^2 - z^2).$$

If we let $r = (x, y, z) \in \mathbb{R}^3$, then $\rho \geq 0$ if and only if $\|r\| \leq 1$. If ρ is a pure state, then $\text{rank}(\rho) = 1$ and $\det(\rho) = 0$, which is equivalent to $\|r\| = 1$. Thus, pure states correspond exactly to the unit sphere in $\mathbb{R}^3$ and mixed states correspond to the open unit ball.

Exercise 1.3.2. What is the pure state $|\psi\rangle$ and its density matrix $|\psi\rangle\langle\psi|$ corresponding to $r_1 = (0, 1, 0)$ and to $r_2 = (0, -1, 0)$.

Solution. By plugging r_1 into Eq. (1.3.1), we get

$$\rho_1 = \frac{1}{2} \begin{pmatrix} 1 & -i \\ i & 1 \end{pmatrix}.$$

We can easily calculate, that the eigenvector corresponding to the eigenvalue 1 of ρ_1 is $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$. By Theorem 1.1.3, we get $\rho_1 = |\psi_1\rangle\langle\psi_1|$. Similarly, we get that $\rho_2 = |\psi_2\rangle\langle\psi_2|$, where $|\psi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$. $\square$

We have the following important bases of pure states that correspond to the intersection points of x, y, z -axes with the Bloch sphere:

- Z -basis: $\{|0\rangle, |1\rangle\}$, $r = (0, 0, 1)$, $-r = (0, 0, -1)$.
- X -basis: $\{|+\rangle, |-\rangle\}$, $r = (1, 0, 0)$, $-r = (-1, 0, 0)$
- Y -basis: $\{|+i\rangle, |-i\rangle\}$, $r = (0, 1, 0)$, $-r = (0, -1, 0)$.

1.4 Measurements

Definition 1.4.1. A *measurement* or a *positive operator valued measure (POVM)* on a Hilbert space $\mathcal{H}$ with a finite set of outcomes Ω is a function $\mu: \Omega \rightarrow \text{PSD}(\mathcal{H})$ such that $\sum_{x \in \Omega} \mu(x) = I$. If $\mu(x)$ are projections, then the measurement is called *projective*.

Axiom 3. If we measure a state ρ using μ , then the probability of getting an outcome $x \in \Omega$ is $p(x) = \text{tr}[\mu(x)\rho]$.

Remark 1.4.2. In general in quantum mechanics, it is not possible to perform measurements on a state without changing it. However, for now, we will not consider what happens with a state after the measurement (we can think that it is destroyed). We only get classical information about which outcome occurred. We will discuss post-measurement states later in the course.

Exercise 1.4.3. Prove that if ρ is a quantum state and μ is a measurement, then $p(x) = \text{tr}[\mu(x)\rho]$ defines a probability distribution on Ω .

Solution. By Theorem 1.1.4(d), we have $\text{tr}[\mu(x)\rho] \geq 0$ since both $\mu(x)$ and ρ are positive. By the linearity of trace,

$$\sum_{x \in \Omega} \text{tr}[\mu(x)\rho] = \text{tr}\left[\sum_{x \in \Omega} \mu(x)\rho\right] = \text{tr}[I\rho] = 1.$$

$\square$

Example 1.4.4 (basis measurement). Let $\{|\psi_i\rangle\}_{i=1}^d$ be an orthonormal basis and let $\mu(i) = |\psi_i\rangle\langle\psi_i|$ be the projection onto $|\psi_i\rangle$. We get

$$p(i) = \text{tr}[|\psi_i\rangle\langle\psi_i|\rho] = \text{tr}[\langle\psi_i|\rho|\psi_i\rangle] = \langle\psi_i|\rho|\psi_i\rangle.$$

In particular, if $\rho = |\phi\rangle\langle\phi|$ is a pure state, then we get

$$p(i) = \langle\psi_i|\phi\rangle\langle\phi|\psi_i\rangle = |\langle\psi_i|\phi\rangle|^2.$$

$\square$

Exercise 1.4.5. What are the outcome probabilities when we measure $|0\rangle$ the X -basis?

Solution. We get $|+\rangle$ with probability $|\langle+|0\rangle|^2 = 1/2$ and $|-\rangle$ with probability $|\langle-|0\rangle|^2 = 1/2$. $\square$

1.5 Bloch Sphere Revisited

Given $r = (x, y, z)$ with $\|r\| = 1$, the positive matrices

$$\mu(0) := \rho(r) = \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix} \quad \text{and} \quad \mu(1) := \rho(-r) = \frac{1}{2} \begin{pmatrix} 1-z & -x+iy \\ -x-iy & 1+z \end{pmatrix}$$

define a general qubit basis measurement. Performing this measurement on a quantum state with Bloch vector $s = (x', y', z')$, then the probability of obtaining the outcome 0 is

$$p(0) = \frac{1}{2} + \frac{1}{2}(r \cdot s) = \frac{1}{2} + \frac{1}{2}(xx' + yy' + zz').$$

Geometrically $r \cdot s$ is the projection of the Bloch vector s of a quantum state onto the axis defining the measurement.

1.6 Uncertainty Relation

Let $|\psi\rangle$ be a pure state. Note that since $X = |+\rangle\langle+| - |-\rangle\langle-|$, we get the following:

$$|\langle\psi|X|\psi\rangle| = |p_X(1) - p_X(-1)| = |2p_X(1) - 1| = 2\max\{p_X(1), p_X(-1)\} - 1,$$

where $p_X(x)$ denotes the probability of getting outcome x after measuring in the X -basis. Clearly,

$$0 \leq |\langle\psi|X|\psi\rangle| \leq 1.$$

The upper bound is attained precisely when either $p_X(1) = 1$ or $p_X(-1) = 1$, that is, when the measurement outcome is certain. The lower bound is attained precisely when $p_X(1) = p_X(-1) = 1/2$, which means that the measurement is completely uncertain. Thus, $|\langle\psi|X|\psi\rangle|$ provides a meaningful way to quantify our uncertainty about the measurement outcome.

Using $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$, we similarly obtain $|\langle\psi|Z|\psi\rangle| = 2\max\{p_Z(1), p_Z(-1)\} - 1$. Combining the two, we get

$$|\langle\psi|X|\psi\rangle| + |\langle\psi|Z|\psi\rangle| \leq 2.$$

Note that by the previous discussion, the equality can never be attained. However, there is a significant strengthening:

Theorem 1.6.1 (Uncertainty relation for Pauli matrices). *For every state $|\psi\rangle$, we have*

$$|\langle\psi|X|\psi\rangle| + |\langle\psi|Z|\psi\rangle| \leq \sqrt{2}.$$

Proof. Let $s_X, s_Z \in \{\pm 1\}$ and let $A = s_X X + s_Z Z$. We need to show that

$$s_X \langle\psi|X|\psi\rangle + s_Z \langle\psi|Z|\psi\rangle = \langle\psi|A|\psi\rangle \leq \sqrt{2}.$$

Using Cauchy-Schwarz inequality, we get

$$\langle\psi|A|\psi\rangle \stackrel{\text{CS}}{\leq} \|A|\psi\rangle\| \leq \|A\|,$$

where $\|A\| = \sup_{\|\psi\|=1} \|A|\psi\rangle\|$ is the operator norm of A . Further,

$$A^\dagger A = A^2 = (s_X X + s_Z Z)(s_X X + s_Z Z) = I + s_X s_Z (XZ + ZX) + I = 2I.$$

This calculation shows that $A/\sqrt{2}$ is unitary. Since the operator norm of a unitary is one, we get

$$\|A\| = \sqrt{2} \|A/\sqrt{2}\| = \sqrt{2}.$$

□

We can interpret the quantity $\max\{p_X(1), p_X(-1)\}$ as the *guessing probability* $p_{\text{guess},X}$, that is, the maximal probability of guessing the outcome of X -basis measurement on the state $|\psi\rangle$ – the best option is to just guess the outcome with larger probability. Using this notation, we can rewrite the uncertainty relation from Theorem 1.6.1 as follows:

$$\begin{aligned} |\langle\psi|X|\psi\rangle| + |\langle\psi|Z|\psi\rangle| &\leq \sqrt{2} \\ 2 \max\{p_X(1), p_X(-1)\} - 1 + 2 \max\{p_Z(1), p_Z(-1)\} - 1 &\leq \sqrt{2} \\ p_{\text{guess},X} + p_{\text{guess},Z} &\leq 1 + \frac{\sqrt{2}}{2}. \end{aligned}$$

So the uncertainty relation from Theorem 1.6.1 gives a bound on the sum of probabilities of guessing the two measurement outcomes correctly.

1.7 Observables

Sometimes it is convenient to reformulate measurements in terms of *observables*. A *projective measurement* is a measurement where all measurement operators are projections, that is, we have a set of outcomes Ω and projections $\{P_x : x \in \Omega\}$ such that $\sum_{x \in \Omega} P_x = I$. Suppose that $\Omega \subseteq \mathbb{R}$. The *observable* associated to the projective measurement is an operator $O \in \text{Lin}(\mathcal{H})$ given by

$$O = \sum_{x \in \Omega} x P_x.$$

Since x are real, O is Hermitian, and, conversely, each Hermitian operator O has a spectral decomposition of this form, thereby defining a projective measurement.

Remark 1.7.1. In physics, observables are often the preferred way to reason about measurements. Measuring the observable O is the same as performing the projective measurement defined by the spectral decomposition of O .

Example 1.7.2. Since $Z = |0\rangle\langle 0| - |1\rangle\langle 1|$, the observable Z corresponds to measuring in the Z -basis with outcomes $\{\pm 1\}$. Similarly, since $X = |+\rangle\langle +| - |-\rangle\langle -|$, measuring X corresponds to measuring in the X -basis with outcomes $\{\pm 1\}$. More generally, we may measure a qubit in the basis given by the antipodal points $r = (x, y, z)$ and $-r$ on the Bloch sphere with outcomes $\{\pm 1\}$. This gives an observable

$$\begin{aligned} O(r) = \rho(r) - \rho(-r) &= \frac{1}{2} \begin{pmatrix} 1+z & x-iy \\ x+iy & 1-z \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1-z & -x+iy \\ -x-iy & 1+z \end{pmatrix} \\ &= \begin{pmatrix} z & x-iy \\ x+iy & -z \end{pmatrix}. \end{aligned}$$

□

An observable provides a compact formula for the expectation value of the measurement outcome. If $O = \sum_{x \in \Omega} x P_x$ is an observable and $\rho \in S(\mathcal{H})$, then

$$\mathbb{E}(\text{outcome}) = \sum_{x \in \Omega} x \Pr(\text{outcome } x) = \text{tr}[O\rho].$$

In case $\rho = |\psi\rangle\langle\psi|$, then the right-hand hand side is equal to $\langle\psi|O|\psi\rangle$.

Chapter 2

Multiple Quantum Systems

2.1 Multiple Systems

If we have classical random variables with outcome sets $\Sigma_1, \dots, \Sigma_n$, then their joint distribution is a probability distribution on the product set

$$\Sigma_1 \times \dots \times \Sigma_n = \{(x_1, \dots, x_n) : x_j \in \Sigma_j\}.$$

So n bits are represented by an n -tuples.

Tensor product is precisely the quantum version of this. If we have bases $\Sigma_1, \dots, \Sigma_n$ for the Hilbert spaces $\mathcal{H}_1, \dots, \mathcal{H}_n$, then

$$\mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_n$$

is the *tensor product* Hilbert space with basis

$$\{|x_1\rangle \otimes |x_2\rangle \otimes \dots \otimes |x_n\rangle : |x_j\rangle \in \Sigma_j\}.$$

We will sometimes use the shorthand $|x_1\rangle|x_2\rangle \dots |x_n\rangle$ or $|x_1 \dots x_n\rangle$. Clearly, we have $\dim(\mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_n) = \dim(\mathcal{H}_1) \dots \dim(\mathcal{H}_n)$ since the basis elements are labelled by the elements of $\Sigma_1 \times \dots \times \Sigma_n$.

Axiom 4. If we have multiple quantum systems with Hilbert spaces $\mathcal{H}_1, \dots, \mathcal{H}_n$, then the joint system has associated Hilbert space $\mathcal{H}_1 \otimes \dots \otimes \mathcal{H}_n$.

Example 2.1.1. If we have two qubits, the joint Hilbert space is $\mathbb{C}^2 \otimes \mathbb{C}^2$, which is 4-dimensional and has the standard basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$.

The Hilbert space of n qubits is $(\mathbb{C}^2)^{\otimes n}$, which is 2^n -dimensional and has the standard basis $\{|x_1 \dots x_n\rangle\}_{x_1, \dots, x_n \in \{0,1\}}$. □